PHY4605 Computational Methods in Physics
Learning Note | Week 7

Numerical Integration: Accumulate a changing quantity and test convergence

Estimate a physically accumulated quantity from sampled values, then test convergence: do refined estimates approach a stable reference?

Physical question. A force changes continuously with time and gradually decays. How can sampled force values estimate the total impulse delivered from $t = 0$ to $t = 2.0\,\mathrm{s}$?

Learning outcomes

1. connect area under a physical curve to an accumulated quantity, limits, and units;
2. trace a trapezoidal estimate and the equivalent MATLAB `trapz(x,y)` calculation; and
3. validate a numerical integral using refinement, an analytic/reference value, and simple error.

How to use this note

Read the physical question and model first. At each prediction checkpoint, pause and write what you expect; then trace the short calculation, run the MATLAB listing, and compare the result with the stated validation evidence.

1. Integration means physical accumulation

In calculus, an integral is often introduced as area under a curve. In computational physics, that area usually represents a physical quantity. For a force that varies with time,

$$J = \int_{t_a}^{t_b} F(t)\,dt$$

is the **impulse** delivered over the stated interval.

The axes carry the unit of the integral. Force is measured in newtons and time in seconds, so

$$\mathrm{N} \times \mathrm{s} = \mathrm{N\,s},$$

which is also equivalent to the momentum unit $\mathrm{kg\,m\,s^{-1}}$. The limits t_a and t_b are part of the physical question: changing them changes what has been accumulated.

Prediction checkpoint. If the force remains positive from 0 to 2.0 s, the impulse must be positive. If the force starts at 12 N and then decays, the impulse must also be smaller than the rectangle $12\,\mathrm{N} \times 2\,\mathrm{s} = 24\,\mathrm{N\,s}$.

2. Reuse a simple decaying force model

Use

$$F(t) = F_0 e^{-t/\tau},$$

with

$$F_0 = 12\,\mathrm{N}, \qquad \tau = 0.8\,\mathrm{s}, \qquad 0 \le t \le 2.0\,\mathrm{s}.$$

The force is largest at the start and decreases smoothly while remaining positive. We want the accumulated impulse

$$J = \int_0^{2.0} F(t)\,dt.$$

For validation later, this teaching model has an analytic integral:

$$J_{\text{exact}} = F_0\tau\left(1 - e^{-T/\tau}\right) = 8.8119840132\,\text{N s}.$$

This value is the independent reference, not the numerical method.

3. A numerical integral only sees samples

A computer does not need the entire continuous curve to apply a basic quadrature rule. In plain language, a quadrature rule estimates accumulated area by combining function values at selected coordinates. Here, **convergence** means that, with the force model and limits held fixed, the numerical estimates approach a stable value as the sample spacing is reduced. With four equal intervals between 0 and 2.0 s,

$$\Delta t = \frac{2.0 - 0}{4} = 0.5\,\text{s},$$

so the five sample coordinates are 0, 0.5, 1.0, 1.5, and 2.0 s.

t (s)	$F(t)$ (N)
0.0	12.0000
0.5	6.4231
1.0	3.4381
1.5	1.8403
2.0	0.9850

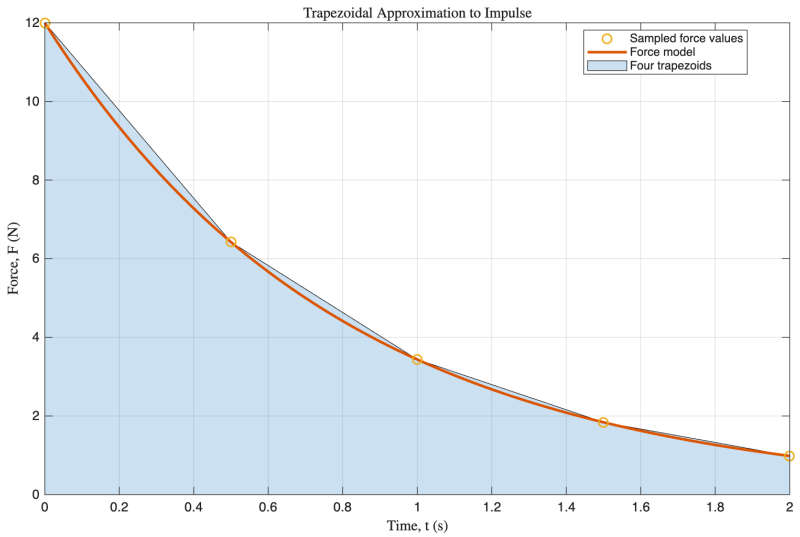


Figure 1: Week 7 force-time model: five samples define four trapezoids whose total area estimates the impulse.

4. One trapezoid is width times average height

Between two neighbouring samples, approximate the curved force by a straight segment. For samples F_i and F_{i+1} separated by Δt,

$$A_i \approx \Delta t \, \frac{F_i + F_{i+1}}{2}.$$

For the first interval, from 0 to 0.5 s,

$$A_1 \approx 0.5 \, \frac{12.0000 + 6.4231}{2} \approx 4.6058 \, \mathrm{N\,s}.$$

The unit is already the unit of impulse because the trapezoid width is in seconds and its height is in newtons.

5. Add the trapezoids for a composite estimate

For uniform spacing, the four trapezoids can be collected into the composite form

$$J_{\mathrm{trap}} = \Delta t \left[\frac{1}{2} F_0 + F_1 + F_2 + F_3 + \frac{1}{2} F_4 \right].$$

Using the five sampled values above,

$$J_{\mathrm{trap}} \approx 9.0969821492 \, \mathrm{N\,s}.$$

This result is positive and below 24 N s, so it passes the basic sign and bound checks. Those checks are useful, but they do not yet tell us whether the approximation is accurate enough.

6. Pseudocode before MATLAB

1. state the accumulated quantity, integration limits, and expected unit;
2. choose a supplied number of intervals;
3. create sample coordinates between the fixed limits;
4. evaluate the force at those coordinates;
5. add trapezoid areas between neighbouring samples;
6. repeat at finer supplied resolutions without changing the force model;
7. compare the estimates with an independent reference; and
8. report the final integral with its unit, error, and physical meaning.

7. MATLAB `trapz` performs the same accumulation

The coarse calculation can be written explicitly and then checked against MATLAB's built-in trapezoidal integration function:

Listing 1: Manual and built-in trapezoidal integration of the sampled force.

```
F0_N = 12;
tau_s = 0.8;
T_s = 2.0;
n = 4;
t_s = linspace(0,T_s,n+1);
F_N = F0_N*exp(-t_s/tau_s);
dt_s = T_s/n;
J_manual_Ns = dt_s*(0.5*F_N(1) + ...
    sum(F_N(2:end-1)) + 0.5*F_N(end));
J_trapz_Ns = trapz(t_s,F_N);
```

The call `trapz(t_s,F_N)` matters: the first argument supplies the physical coordinates and therefore the actual spacing between samples. The manual and `trapz` estimates should agree for the same data.

Runnable does not mean physically correct. The call `trapz(F_N)` runs, but MATLAB then assumes unit spacing between samples. If the real sample spacing is 0.5 s, reporting that raw result directly as an impulse silently loses the physical time scale.

8. Refine the sampling without changing the physics

Now keep F_0, τ, and the integration limits fixed while increasing only the number of intervals. This isolates the effect of numerical resolution.

For each row, the **absolute error** is the size of the difference from the analytic reference,

$$E_{\text{abs}} = |J_{\text{trap}} - J_{\text{exact}}|,$$

while the **relative error** expresses that difference as a percentage of the reference,

$$E_{\text{rel}} = \frac{E_{\text{abs}}}{|J_{\text{exact}}|} \times 100\%.$$

Intervals	Δt (s)	J_{trap} (N s)	absolute error (N s)	relative error (%)
4	0.500	9.0969821	0.2849981	3.2342
8	0.250	8.8835797	0.0715957	0.8125
16	0.125	8.8299047	0.0179207	0.2034

The estimates move toward the same analytic reference as the sample spacing is reduced. For this smooth model and these supplied resolutions, the error also decreases each time the spacing is halved.

9. Required validation: refinement plus reference comparison

The finest supplied estimate is

$$J_{16} = 8.8299047499\,\text{N s}.$$

Compared with the analytic reference,

$$|J_{16} - J_{\text{exact}}| = 0.0179207367\,\text{N s},$$

and the relative error is approximately

$$0.203\%.$$

Validation statement. The numerical impulse remains positive and below the physical upper bound, and the trapezoidal estimate approaches the analytic result as the sampling is refined. At 16 intervals the result differs from the analytic reference by about 0.203%, supporting the estimate for the stated model and limits.

10. Interpret the number physically

An impulse of about $8.83\,\mathrm{N\,s}$ corresponds to a momentum change of about $8.83\,\mathrm{kg\,m\,s^{-1}}$ in the positive force direction over 0 to 2.0 s. This interpretation is conditional on the stated force model and limits; numerical precision cannot repair an inappropriate model.

Guided extension: Simpson comparison and convergence evidence

Read and interpret the supplied eight-interval Simpson result, $J_{\mathrm{Simpson}} = 8.8124455163\,\mathrm{N\,s}$, as comparison evidence; independent derivation or implementation of Simpson's rule is not required. In the trapezoidal table, halving Δt reduces error by about a factor of four; describe that observed convergence pattern without deriving its formal order.

Optional extension

Optional extensions are quadrature error-order derivation, independent composite-rule comparison, and adaptive quadrature. They are not required for this week's pathway.

Three takeaways

1. A numerical integral represents physical accumulation over explicit limits, and its unit combines the two axis units.
2. The trapezoidal rule sums areas between neighbouring samples; `trapz(x,y)` uses their supplied coordinates.
3. Refine the sampling and compare with a reference before trusting the accumulated result.

Preparation for the practical. Transfer the same reasoning to impulse, work, and charge. For each context, identify the accumulated quantity, limits, axis units, trapezoidal sampling/refinement plan, and validation reference.